

Julio Villalvazo

Full-Stack Developer | Java, Spring Boot, React, TypeScript AWS, GCP, AI/ML Integrations

Full-Stack Software Engineer with 4+ years of experience building scalable backend services, modern frontend applications, cloud-native systems, and AI-powered automation solutions. Currently working directly with Amazon Audible on large-scale audiobook, audio-text synchronization, content moderation, and pre-ingestion platforms using Java, Spring Framework, React, TypeScript, AWS, DynamoDB, S3, Bedrock, CDK, EC2, Playwright, Vitest, and internal CI/CD pipelines. Also working with TCS for Visa on fraud detection systems involving Java-based microservices, Kafka messaging, Kubernetes pods, Cassandra configuration, secure deployment values, and GCP infrastructure. Previous experience delivering high-impact solutions for Google/Alphabet, USAA, Imagine Learning, and Cyberpuerta. AWS Certified Developer Associate with hands-on experience in backend engineering, frontend architecture, cloud infrastructure, DevOps, testing strategy, API platforms, and AI/ML integrations.

WORK EXPERIENCE

FullStack Developer

June 2025 – Present

Amazon Audible, Mexico City

- Built configurable filter profiles using DynamoDB to enable or disable AI validation rules during the audiobook pre-ingestion process across multiple environments.
- Designed and implemented AI-powered pre-ingestion filters for audiobook cover art validation using AWS Bedrock, DynamoDB, S3, and internal content moderation workflows.
- Implemented prompt versioning and auditability by storing AI prompt history in Amazon S3, improving traceability, governance, and compliance for AI-based content validation.
- Integrated AWS Bedrock prompt caching to reduce repeated prompt processing, improving request performance and lowering token consumption from approximately 12,000 tokens per filter run.
- Designed and configured integration tests in CI/CD pipelines across multiple deployment stages, including creation and validation of the required AWS resources for each environment.
- Designed and Implemented an AI-based hate imagery and global content moderation service for audiobook cover art, supporting marketplace-specific blocking rules based on detected moderation categories.
- Designed DynamoDB-based marketplace mapping rules to determine whether specific AI-detected hate imagery or sensitive content should block audiobook publication in each marketplace.
- Developed and maintained a custom audio-text synchronization platform using React, TypeScript, Java, Spring Framework, AWS, DynamoDB, S3, and EC2.
- Improved a highly customizable transcript synchronization UI with features such as word highlighting, narrated-word tracking, custom hotkeys, configurable highlight colors, timing metadata, and word position data.
- Built frontend functionality using React hooks, Zustand, TypeScript, Tailwind CSS, Material UI, and component-based architecture, improving maintainability and development speed compared with the previous Redux-based approach.
- Planned and implemented the frontend testing strategy using Vitest for unit testing and Playwright for visual regression testing.
- Supported end-to-end validation using Selenium-based internal Amazon testing frameworks.
- Currently designing the migration of application deployment pipelines from MAWS to NAWS architecture, enabling infrastructure orchestration through AWS CDK and improving deployment speed, maintainability, and environment consistency.

CONTACT

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CORE STRENGTHS

Full-stack software engineering with strong ownership across frontend, backend, cloud, testing, and deployment workflows.

Experience building enterprise systems for large-scale companies including Amazon Audible, Visa, Google/Alphabet, USAA, and Imagine Learning.

Hands-on experience with AI-powered automation, content moderation, fraud detection, audio-text synchronization, API platforms, and cloud-native applications.

Strong ability to design maintainable React applications using TypeScript, hooks, Zustand, Material UI, Tailwind CSS, testing frameworks, and component-based architecture.

Backend experience with Java, Spring Boot, Spring Framework, Kotlin, Micronaut, microservices, REST APIs, Kafka, Cassandra, DynamoDB, and distributed systems.

Cloud and DevOps experience with AWS, GCP, Kubernetes, Docker, CI/CD, AWS CDK, infrastructure configuration, and deployment automation.

Testing ownership across unit tests, integration tests, E2E tests, visual regression testing, and CI/CD pipeline validation.

SKILLS

Backend: Java, Spring Boot, Spring Framework, Kotlin, Micronaut, Node.js, Express.js, REST APIs, Microservices, API Design,

FullStack Developer

Tata Consultancy Services (TCS), Visa Client, Remote

April 2026 – Present

- Working on a fraud detection platform using Java-based backend services, distributed systems, cloud infrastructure, and secure deployment configuration.
- Configuring and maintaining infrastructure values for enterprise services, including Cassandra IPs, deployment configuration files, secure values, scaling parameters, and runtime environment variables.
- Supporting DevOps and cloud infrastructure activities using Google Cloud Platform (GCP) for production environments.
- Working with interconnected applications that communicate through Apache Kafka messaging, supporting event-driven processing for fraud detection workflows.
- Managing and updating Kubernetes pod configurations for development and production environments.
- Using internal deployment tooling for development application configuration and GCP-based deployment workflows for production application environments.
- Collaborating on scaling, configuration management, secure value handling, and deployment readiness for Java microservices in a distributed fraud detection ecosystem.

FullStack Developer

Tata Consultancy Services (TCS), USAA Client, Remote

September 2024 – December 2025

- Developed internal company web platform using Java with Spring Boot for backend services and React with Material UI and other modern libraries for the frontend.
- Implemented unit testing in Java for backend logic and in React using modern testing tools.
- Contributed to the modernization of USAA's internal systems by developing scalable, maintainable code following agile practices.

FullStack Developer

Tata Consultancy Services (TCS), Alphabet (Google) Client, Remote

April 2024 – September 2024

- Collaborated with Google Cloud Platform (GCP) teams to implement and optimize API management solutions using ApigeeX.
- Designed, deployed, and secured APIs with custom policies, traffic management strategies, and analytics to support high-scale environments.
- Automated deployment workflows and contributed to internal tooling that streamlined API proxy lifecycle management.
- Gained hands-on experience in one of the world's most advanced cloud ecosystems, reinforcing best practices in scalable and secure API design.

FullStack Developer

Encora, Imagine Learning, Remote

May 2024 – Present

- Led migration of a core exam-creation service from C#/.NET to Kotlin using Micronaut, enhancing performance and maintainability.
- Maintained and optimized company platforms by implementing new features and resolving issues using React, TypeScript, and Storybook.
- Developed reusable, modular components adhering to design standards from the UX/UI team, with thorough testing in Storybook to ensure consistency and reliability.

Frontend Developer

Cyberpuerta, Remote

March 2023 – April 2024

- Developed onboarding process for new app installs, including connection to Firebase, and petitions to the api using React-Native and react-query.
- Improved the app performance, using different React hooks, contexts and libraries (reassure, flipper, React dev tools profiler, etc).

Distributed Systems, Event-Driven Architecture

Frontend: React, TypeScript, JavaScript, React Hooks, Zustand, Redux, Material UI, Tailwind CSS, HTML, CSS, Responsive UI, Component-Based Architecture

Cloud & DevOps: AWS, Google Cloud Platform, AWS Bedrock, AWS CDK, DynamoDB, S3, EC2, Kubernetes, Docker, CI/CD Pipelines, Infrastructure as Code, Cloud Deployments, Environment Configuration

Data & Messaging: DynamoDB, Cassandra, PostgreSQL, MySQL, MongoDB, GraphQL, Apache Kafka, Event Streaming, Data Modeling, SQS, SNS

Testing & Quality: Unit Testing, Integration Testing, End-to-End Testing, Visual Regression Testing, Vitest, Playwright, Selenium, Cypress, Jest, Storybook, Chromatic

AI / Machine Learning

Integrations: AWS Bedrock, AI Content Moderation, Prompt Engineering, Prompt Caching, Prompt Versioning, AI Governance, AI-Based Validation Workflows

API & Security: ApigeeX, API Gateway Concepts, Secure API Design, Authentication, Authorization, API Policies, Traffic Management, Content Moderation, Compliance-Oriented Development

Tools: Git, GitHub, GitLab, DBeaver, Firebase, React Query, Expo, Flipper, React DevTools Profiler

Languages:

- Spanish native
- English fluent
- Japanese beginner

OTHER

- Backend Internship as International Experience (Python, Docker, MySql) in AlaTest, Stockholm, Sweden.
- Volkswagen Internship as telemetry engineer, designed an

- Created over 120 components following the design provided by the UX/UI team, using Typescript, react-native-paper and tailwind, and tested them using Jest.
- Created over 120 stories using Chromatic Storybook.

FullStack Internship

November 2022 – March 2023

Cogent IBS, Remote

- Developed multiple solutions for different frontend and backend issues using Typescript, Javascript and React, solving in total 135 issues as a team.
- Designed two new features for an internal use web application (employees visa tracking and employees visa tracking registration).
- Reported in daily reunions about work in progress and estimated delivery time.
- Managed databases using MySQL and PostgreSQL with DBeaver.
- Deployed CRUD queries for Lambdas based in AWS using Node.js and Database maintenance based in AWS.

Embedded Software Engineer

July 2021 – Present

Tsool Technology, Orizaba, Veracruz

- Co-founded Tsool Technology to develop solutions integrating hardware and software.
- Planned projects using SCRUM methodology and organized research data.
- Conditioned Raspberry Pis (3B+, 3A, Zero W).
- Managed databases using SQL and GraphQL.
- Created enclosing cases for hardware using CAD software (SolidWorks).

Electric Maintenance Engineer

July 2020 – July 2021

Kimberly Clark de Mexico, Apizaco, Tlaxcala

- Conditioned and maintained ac and dc motors for different voltages in a range from 120V to 480V.
- Installed and monitored electric sensors using Rockwell PLCs (ControlLogix, MicroLogix, CompactLogix, Input I/O, Compact I/O) .
- Programmed and maintained rockwell PLCs using RSLogix and RSLogix5000.
- Managed databases using SQL.
- Managed an electric maintenance team, achieving installation for an extruder heating system, safety sensors and guards installations, motor maintenance.
- Programmed Rockwell Variable Frequency Drivers for ac and dc motors.

EDUCATION

Masters in Cybersecurity

August 2024 - Present

Instituto Tecnológico y de Estudios Superiores Monterrey (ITESM)

Currently studying

Bachelor of Science

December 2020

Instituto Tecnológico y de Estudios Superiores Monterrey (ITESM)

Major in Mechatronics Engineering

HMI for an autonomous vehicle using Python.

- Studied Italian in Rome for 3 months
- SolidWorks certification
- LabView certification
- Volunteered in a 6-month project for a disabled people association designing a low cost wheelchair
- Japanese Oratory Contest winner in 2012